

Project Proposal

CSE 4204

Northern University of Business and Technology, Khulna
Department of Computer Science and Engineering

1. Team Information

Field	Information
Team Number	T03
Official Team Name	CSE4204-8B-T03
Section	8B
Project Title	AI Study and Task Planner
GitHub Repository Link	https://github.com/saimhasan10/CSE4204-8B-t03-ai-study-and-task-planner

Role	Member Name	Student ID
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2. Project Title

AI Study and Task Planner

3. Problem Statement

University students often struggle to manage study tasks, project deadlines, exam preparation, and regular academic responsibilities at the same time. Many students keep tasks in different places such as notebooks, messaging apps, calendars, or memory, which makes it easy to forget important deadlines and creates last-minute pressure.

Existing task management tools can store tasks and schedules, but they are usually general-purpose systems. They do not properly understand student-specific needs such as exam preparation, lab work, project milestones, and priority-based study planning. Most of them also do not provide intelligent guidance for breaking large tasks into smaller steps or creating a practical study plan based on deadlines.

AI is relevant for this problem because it can analyze the student's tasks, deadlines, priority levels, and available time, then suggest a simple plan. The system will help students decide what to do first, how to divide large tasks, and how to stay organized during academic pressure.

4. Objectives

- Build a simple and useful planner for university students.
- Help students organize study tasks, project work, and deadlines in one place.
- Generate AI-based study plans according to task priority and deadline.
- Break large tasks into smaller actionable steps using AI support.
- Improve student productivity and reduce last-minute pressure.
- Develop a complete full-stack AI-based software system within the semester timeline.

5. Proposed Features

- User registration and login system

- Student dashboard with task summary and progress view
- Task management with title, description, deadline, priority, and status
- AI study plan generator based on tasks, deadlines, and priority
- AI chatbot for study planning and task breakdown guidance
- Saved AI plans and task progress tracking
- Clean responsive interface for desktop and mobile users

6. Technology Stack

Part	Technology
Frontend	React.js, Tailwind CSS
Backend	Django, Django REST Framework
Database	PostgreSQL
AI Integration	Gemini API / OpenAI API
Version Control	GitHub
Deployment Plan	Vercel for frontend, Render/Railway for backend, Supabase/Neon for PostgreSQL

7. AI Integration Plan

The AI feature will be integrated through the backend, not directly from the frontend. The React frontend will send user tasks and questions to the Django REST API. The Django backend will collect task information from PostgreSQL, prepare a clear prompt, send it to the AI service, receive the AI response, save the result in the database, and return it to the frontend.

- AI will generate weekly or daily study plans from the user's tasks and deadlines.
- AI will suggest task priority based on urgency and importance.
- AI will break large tasks into smaller steps.
- AI chatbot will answer simple study-planning questions using the user's task context.

8. Expected Outcome

After using this system, students will be able to manage their study tasks, deadlines, and project work in one organized place. They will get AI-generated planning support that helps them understand which task should be completed first and how to divide large tasks into smaller steps.

The expected benefit is better time management, reduced deadline pressure, improved productivity, and a more structured academic workflow. The system will be simple enough for regular students to use and practical enough to demonstrate meaningful AI integration.

9. Similar Existing Systems and Limitations

Existing System	Main Use	Limitation for Students
Todoist	Task and deadline management	General-purpose task manager; limited student-focused AI planning
Google Calendar	Scheduling and reminders	Good for dates, but not focused on academic task breakdown
Trello	Board-based task tracking	Useful for teams, but does not automatically generate study plans
Notion	Notes and productivity workspace	Flexible, but planning depends heavily on manual setup
Student planner apps	Academic task planning	Often lack AI-based priority suggestions and task breakdown

10. Initial Project Planning

Timeline	Planned Work
Week 01	Team formation, idea selection, and initial research
Week 02	Project proposal, team information sheet, GitHub repository setup, and initial planning
Week 03	SRS preparation with requirements, user roles, and use cases
Week 04	System design: architecture, ER diagram, use case diagram, and API structure
Week 05	UI/UX wireframe design using Figma
Week 06	Backend development: authentication, database connection, and REST APIs
Week 07	Frontend development: dashboard, forms, navigation, and API connection
Week 08	AI integration: chatbot and AI study plan generator
Week 09	Feature completion and improvement
Week 10	Testing and debugging
Week 11	Deployment of frontend, backend, and database
Week 12	Final documentation, user manual, and API documentation
Week 13	Presentation preparation and demo practice
Week 14	Final presentation and viva

11. Feasibility and Scope Control

The project scope is intentionally kept realistic for one semester. The team will focus on core features first: authentication, task management, AI plan generation, progress tracking, and a clean dashboard. Complex features such as calendar synchronization, mobile app development, payment system, or social networking features will not be included in the first version.

12. Conclusion

The AI Study and Task Planner will help students manage their study tasks, deadlines, and project work in a more organized way. By using AI, the system will generate study plans, suggest task priorities, and break large tasks into smaller steps. This will improve student productivity and reduce last-minute pressure. Overall, the project is simple, realistic, and useful for university students.